



European Southern Observatory

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Detectability of the Habitable Zone Inner-Edge Discontinuity with PLATO

Bioverse forecasts for the PLATO target samples

Master Thesis

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Abstract

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1. Introduction

1.1. Motivation

Whether Earth-like conditions on other planets in our galaxy are common is one central question in modern astronomy, which observational data itself has not been able to answer yet. The discovery of thousands of exoplanets in the last two decades have shown that small planets are ubiquitous around all types of dwarf stars [1, 2], with an occurrence-rate distribution that is bimodal across the Fulton radius valley near $1.8 R_{\oplus}$ separating rocky super-Earths from sub-Neptunes [3], and that $\eta_{\oplus} \sim 0.2\text{--}0.6$ rocky planets sit in the classically-defined habitable zone of each solar-like host star [4, 5]. Yet the observational sample of planets receiving near-Earth irradiation remains sparse, biased toward large radii, and with the exception of a small number of bright nearby M dwarfs dominated by hosts, too faint for the atmospheric follow-up that a habitability claim would require [6, 7]. The habitable zone therefore remains, more than three decades after its modern formulation [8, 9], a theoretical construct grounded on essentially a single data point.

A clear illustration of this comes from Venus and Earth. Both accreted from overlapping mass reservoirs [10, 11], exhibit bulk densities within a few percent of each other [12], and are thought to have undergone similar formation stages. However, their current states differ dramatically. Whether Venus lost its water content in the past still remains open [13, 14].

The runaway greenhouse effect is thought to separate them: If a planet's irradiation exceeds the threshold, where it is higher than the outgoing longwave radiation, no stable state of liquid surface water is possible and the entire oceans evaporate into a steam atmosphere, leading to a net radius inflation. This observation, that the runaway transition is imprinted on the transit radius reframes the question of habitable zone. Instead of attempting to prove per-planet if it is habitable, we aim to ask whether the whole *population* of small planets this step in mean transit radius. A statistical detection of this discontinuity would provide a first observational test of the habitable zone concept itself, whereas a non-detection would deliver an upper limit on water content in the population of small planets [6, 15, 16]. Schlecker et al. [17] formalised this idea into a concrete predictive framework, which terming the habitable-zone inner edge discontinuity, or HZIED.

1.2. Aims and contributions of this thesis

This thesis extends the HZIED framework of Schlecker et al. [17] along with an evaluation on actual PLATO targets.

Real target samples. The planet population generator ingests the PLATO Input Catalogue directly and produces planet catalogues for the P4 (M-dwarf, $V \leq 16$) and P5 (FGK, $V \leq 13$)

samples defined by [Nascimbeni et al. \[18\]](#). Per-sample radius precisions are adopted from the *PLATO Definition Study Report [19]* rather than assumed to be uniform across the survey.

Real stellar ages Where the original framework drew stellar ages from a uniform prior on $[0, 10]$ Gyr, this thesis uses the Gaia DR3 FLAME isochrone catalogue [\[20\]](#) directly, or a bootstrap fill from the FLAME distribution when no FLAME value is available, or a blend of FLAME plus a uniform prior for the unmeasured tail.

Physically extended steam prescriptions. Beyond the [Turbet, Bolmont, Ehrenreich, et al. \[21\]](#) grey-atmosphere table used in the original framework, additional prescriptions are compared side by side: a AGNI radiative-convective grid, an AGNI+PALEOS-updated version of that grid, the [Aguichine et al. \[22\]](#) irradiated-ocean model, the **Pierrehumbert2023** Hycean prescription, and a mixed rocky/Hycean population motivated by [Innes, Tsai, and Pierrehumbert \[23\]](#).

The overarching question addressed by these extensions is:

Can, and if so, when during mission time can PLATO detect the population-level radius discontinuity predicted by the runaway greenhouse transition, and which combination of stellar age distribution, planetary composition, and observed target sample is required to make the detection decisive?

1.3. The habitable zone and its inner edge

1.4. The runaway greenhouse and its observable signature

1.5. A demographic test: the habitable-zone inner-edge discontinuity

1.6. The PLATO mission

1.7. Contributions of this thesis

1.8. Structure of this thesis

2. The Habitable Zone and the HZIED Test

We now aim to setup the physics and observational vocabulary for this work. [Section 2.1](#) defines the classical habitable zone and locates its inner edge at the runaway greenhouse threshold. [Section 2.3](#) covers the transit method and the precision it delivers on radii of small planets. [Section 2.4](#) sketches the small-planet population. [Section 2.2](#) closes by defining the HZIED test itself, the specific hypothesis that [Schlecker et al. \[17\]](#) proposed and this thesis extends.

2.1. Habitable Zone

With the habitable zone we refer to the circumstellar region where a earth-like rocky planet (holding a $\text{CO}_2/\text{H}_2\text{O}/\text{N}_2$ atmosphere regulated by a functioning carbonate–silicate cycle) can sustain an amount of liquid surface water [\[9, 24\]](#) *[?] Is this definition correct?*. It is bounded by the inner edge where a water-saturated atmosphere can no longer radiate the incoming stellar flux (instellation) back to space, and the runaway greenhouse effect sets in. The outer side is bounded by the classical maximum-greenhouse limit of [Kasting, Whitmire, and Reynolds \[24\]](#) and [Kopparapu, Ramirez, Kasting, et al. \[9\]](#):

CO_2 condenses out of the atmosphere and the greenhouse effect is no longer sufficient to keep the surface temperature above the freezing point of water [\[9\]](#). Both boundaries, functions of stellar type and atmospheric composition, are theoretical rather than observed. They emerge from 1D radiative-convective *[?] what is this exactly; worth mentioned here?* climate models assuming earth-like compositions.

Unlike the inner edge, the outer edge is not a sharp threshold but rather a gradual transition from habitable to uninhabitable conditions. It can be extended by non-condensing greenhouse gases: primordial H_2 –He mixtures accreted from protoplanetary disks can prevent the surface from freezing [\[25\]](#), while continuous volcanic outgassing of H_2 pushes the classical outer edge to ~ 2.4 AU in the solar system [\[26\]](#). Volcanic CH_4 can similarly widen the outer HZ by $\sim 8\%$ for a solar-analog host, though it cools rather than warms atmospheres around cooler M-dwarfs (see [Ramirez \[27\]](#) for a review).

Most importantly for this thesis, the outer transition leaves no clean signature on the transit-radius plane. A cold rocky planet near the maximum-greenhouse limit has essentially the same bulk radius as its temperate counterpart: no atmospheric phase transition alters its envelope mass or bulk structure in a detectable way. The runaway greenhouse, by contrast, inflates transit radii by tens of percent ([Section 2.2](#)), producing a step function in the population-averaged radius distribution. The HZIED test therefore focuses exclusively on the inner edge, and the

outer boundary appears in the rest of this thesis only as context.

2.2. The Habitable Zone Inner-Edge Discontinuity

The HZIED is the statistical signature of the runaway greenhouse effect on the transit-radius of a population of small planets. This claim could be divided into three lines: (i) a small rocky planet with an amount of surface water crossing the runaway threshold S_{thresh} inflates its transit-observable radius by tens of percent ; (ii) at a fixed water inventory, the change of radius reveals a step function in instellation S and (iii) this step function survives averaging over an entire population of rocky planets despite noisy detection. The detection of the HZIED would therefore be a strong evidence for the runaway greenhouse transition on exoplanets.

Three physical ingredients had to be assembled before the idea became a testable prediction. [Turbet et al. \[15\]](#) first pointed out that a steam-dominated atmosphere on a runaway planet is transit-observable, with a radius inflation of tens of percent for Earth-mass planets. [Turbet, Bolmont, Ehrenreich, et al. \[21\]](#) then computed a full mass–radius grid for rocky planets more irradiated than the runaway greenhouse limit, tabulating the inflated transit radius as a function of planet mass and water mass fraction $x_{\text{H}_2\text{O}}$. [Dorn and Lichtenberg \[28\]](#) added the constraint that a substantial fraction of the outgassed water dissolves back into the silicate melt during the magma-ocean phase and does not contribute to the atmospheric expansion, reducing the effective radius inflation by up to a factor of two at $x_{\text{H}_2\text{O}} \sim 10^{-2}$. Together these three form the “Turbet+2020 with Dorn–Lichtenberg (DL) correction”, forming the original HZIED framework.

2.3. Transit Method

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A. Additional Material

Additional derivations, validation plots or tables can be placed here.

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